

Part 2. Machine Learning

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1. Introduction

Providing a company with comprehensive cybersecurity practices and methods is crucial to its overall protection and livelihood. ScottishGlen have been recognised as a target for a cyberattack by a hackivist group, meaning evaluating ScottishGlen's resilience against an attack is critically important. An assessment of ScottishGlen's network traffic alluded to suspicious activity and a classifier using machine learning to categorise the network packet data should be developed.

The technical staff of ScottishGlen provided a network traffic dataset containing data for various packets which contained headers covering different aspects of the packet, including the attack category for the packet. Identifying which attack categories are actually attacks and which are benign is critically important to develop the classifier and machine learning algorithms.

This report will suggest 2 different machine learning algorithms, describe how each of them work in the network packet context, how to implement each algorithm and evaluate their effectiveness for the network.

Machine learning

Machine learning is the act of using algorithms which improve based upon training data the algorithms are continuously fed. The machine learning algorithms should be able to dynamically learn from previous data and therefore be able to predict certain outcomes based upon the training and finetuning the algorithm undergoes. Machine learning exists in many different contexts such as for facial recognition, email spam filtering and stock market prediction. If there's a situation where data can help assist in predicting what will occur, machine learning could be present. The type of machine learning technique that is used for identifying trends is known as regression and is essential to predictive modelling (Seldon, 2021). Another type of machine learning are classifiers which aim to categorise data into certain classes based on predictive outputs (Mesevage, 2020) and will be used for the datasets provided by the technical team. Machine learning within cybersecurity can be found within automated penetration testing, anomaly detection and intrusion detection systems. These applications of machine learning can assist blue team and red team operations within a company while also reducing man power but not risking accuracy being affected, assuming the model and has been sufficiently trained.

IDS and NIDS

Intrusion detection system (IDS) is a piece of software which aims to identify any suspicious activity found within the device it is protecting. What qualifies as suspicious is based upon policies and parameters an administrator would set. If a policy was broken, the details and attributes of what caused the detection to spike would be saved and the administrator would be notified. A variation of IDS would Network Intrusion detection system (NIDS) which has a focus on the network and analyses the network traffic being received or within the network (Barracuda, n.d.). The output provided in 'CMP417_training dataset1' is from a NIDS.

Detecting the potentially suspicious or malicious activity is most commonly either signature based or anomaly based. Signature based detection utilises past knowledge of specific byte sequences or instructions which would commonly be used within network attacks or malware. Signature based is very effective in identifying pre-existing attacks (assuming the IDS is configured with the knowledge of how to detect them) but will be unable to identify completely unique and new attacks due to no signature for it existing. Comparitively anomaly based systems utilise newer technology (such as machine learning) to identify the difference between accepted, trustworthy activity and potentially

malicious activity. If the activity does not follow the predicted outcome created by machine learning, this could be viewed as suspicious and the administrators would be alerted. While this is effective in identifying unknown threats, it can lead to many false positives. A similar software to IDS would be Intrusion Prevention Systems (IPS), where IDS aims to passively monitor and alert a technical team of suspicious activity, an IPS would aim to immediately stop the potential attack from occurring (Paloalto, n.d.). While this is effective at stopping damage from occurring within the network, it results in false positives having a greater impact due to the harmless activity being completely halted.

2. Machine learning algorithms

To most accurately predict when an attack has occurred, the machine learning algorithm being chosen must be carefully chosen. The machine learning algorithms suggested for ScottishGlen to implement are Random Forest and Naïve Bayes.

Random forest

The core functionality of random forest comes from another machine learning algorithm known as decision trees where multiple decision trees are constructed to train the model to the predicted outcome. Decision trees start at the root of a tree and branch out based upon the features of present within the data source with the end (or final node) acting as the final prediction for the tree. For example, within the dataset provided the attributes for where the nodes could branch out could be for the protocol, service, duration, and attack category. A rough example of what a classifier for the dataset provided in 'CMP417_training dataset1' could look like:

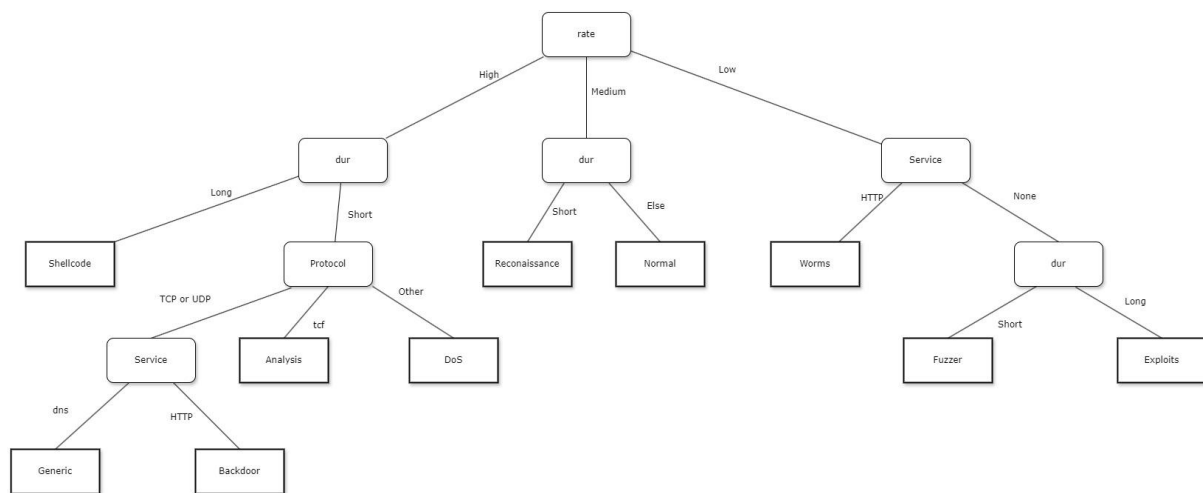


Figure 1. A rough example of a classifier which could be used for the dataset.

Different types of decision tree algorithms exist which can have different effects on the number of sub-nodes or how often they split off. Additionally, the amount of entropy and information gain can vary. Entropy refers to the amount of impurity or uncertainty found within a node, meaning a high entropy can result in a larger variance of results and low entropy means that the class labels are quite homogenous (Seth, 2023). The entropy of a given node can be calculated using the equation:

$$Entropy(n) = - \sum_i p(i|n) \log p(i|n)$$

Where: $p(i|n)$ is the relative frequency of class i at node n

Information gain draws connections to entropy by measuring the decrease of entropy by splitting the dataset (Brownless, 2020). Maximising information gain for each node should be prioritised. Information gain of a node can be calculated using the equation:

$$\text{Information Gain} = \text{Entropy}(p) - \left(\sum_{i=1}^k \frac{n_i}{n} \text{Entropy}(i) \right)$$

Where: *parent node (p)* is split into *k* partitions, and *n_i* is the number of records in partition *i*

With multiple decision trees, the random forest algorithm can begin to take shape. The algorithm will use ensemble methods in a way to integrate all the previous decision trees to generate a more effective predictive model. The majority vote from the various decision trees (which have all been given the same dataset) is taken via averaging which results within a final prediction and a more accurate, in this case classifier machine learning algorithm (IBM, n.d.).

Improving the random forest model can be tackled in numerous ways such as altering certain hyperparameters or the quantity of decision trees within the random forest. To improve the predictive power (more accurate result) adjusting the `max_features`, which is the highest number that the random forest can split a node, resulting in a balance between complexity and generalisation. Improving the speed of the model can be achieved with the `n_jobs` hyperparameter which affects the number of processors the engine generating the forest is allowed to use. Alternatively, a solution not tied to computer hardware would be `random_state` which is responsible for controlling the randomness of final ‘votes’ of each decision tree. Reducing the forest error rate can be achieved by increasing the amount of decision trees within the random forest.

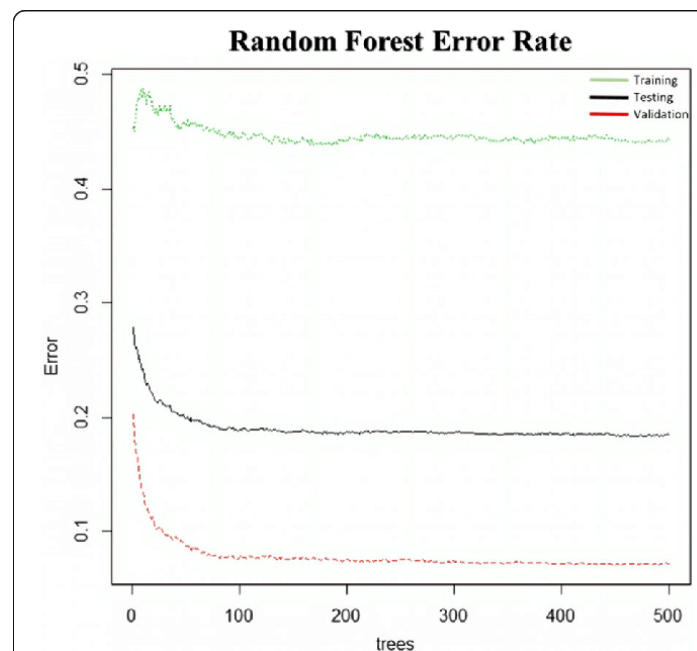


Figure 2. Example of random forest error rate decreasing logarithmically as the amount of decision trees increases (Darshini, 2024).

To perform data cleaning, the various decision trees should be ‘pruned’ which involves getting rid of tree attributes or nodes which do not assist in classifying (Hoare, 2024). However, the random forest voting system is effective in identifying outliers from the predictive outcome. Regarding modelling, utilising both the provided training and testing datasets to verify the classifier fits the model.

Naïve Bayes

The naïve Bayes machine learning algorithm is firstly derived from Baye's theorem which aims to use already gathered information about the problem to use it to help develop a more accurate hypothesis (Ray, 2024). Baye's theorem can be represented as the equation:

$$P(h|d) = (P(d|h) * P(h)) / P(d)$$

Where: $P(h|d)$ is the posterior probability, $P(d|h)$ represents the probability that data (d) is true if hypothesis h is true, $P(h)$ is the prior probability and $P(d)$ is the probably for the data.

Due to machine learning being centred around wanting to correctly predict outcomes based on data, maximising the hypothesis based on training data used within a classifier example results in the most effective outcome for a classifier machine learning example.

What makes Naïve Bayes "naïve" comes from how during the classification process where the algorithm assumes that features within the dataset are independent and therefore other features do not have any effect on each other, which works for this situation where the different attributes of the network packet will not influence the other. This results in the calculation for the probabilities of each hypothesis being much more straightforward, this results in the algorithm being extremely efficient and low maintenance for a computer while processing the predicted hypothesis's, meaning that naïve Bayes is also incredibly useful for big datasets. Additionally, naïve Bayes can work as either a binary classifier or a multi-class classifier, which would make it very suitable for the processing and training from the NIDS dataset (Brownlee, 2020).

Different types of Naïve Bayes exist, such as Gaussian Naïve Bayes which uses Gaussian distribution (also referred to as normal distribution) to distribute the data effectively and easily since only the mean and standard deviation of the input data is required for each class. Having your input data act as closely to gaussian distribution is ideal, therefore when data cleaning removing elements which fall out of the gaussian distribution is recommended. Likewise with the random forest algorithm, utilising both the training and testing datasets is essential for modelling.

3. Analysis and evaluation of the suggested algorithms

Evaluating if the algorithms suggested for the technical staff to implement is critical to verify if they were effective in acting as a classifier for calculating if a packet was an attack based on the NIDS dataset. To achieve this, using model performance metrics are suggested. One of these would be a confusion matrix, which is used to differ if a result is correctly giving the expected output. The predicted output refers to what the machine learning algorithm predicted.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Figure 3. Example of a confusion matrix (Narkhede, 2018).

The confusion matrix can be interpreted as follows:

- True positive refers to when the predicted output is positive, and the actual output is positive as well.
- True negative refers to when the predicted output is negative, and the actual output is negative as well.
- False positive refers to when predicted output is positive, however the output should be negative.
- False negative refers to the when the predicted out is negative, however the output should be positive.

NIDS will very commonly flag packets incorrectly, creating false positives, resulting in alerts being created when they should not be.

Using the confusion matrix, it is possible to evaluate the effectiveness of the algorithms by calculating certain metrics such as the recall, precision, accuracy, and F-measure (Narkhede, 2018).

- Recall is defined by the number of positive examples; this value should be high if the result is better.
- Precision refers to the accuracy of positive cases.
- Accuracy is the measure of how many have been correctly predicted.
- F-measure considers the recall and precision of models with a high F-measure resulting in good precision and recall.

To conclude, the technical team should use the suggested machine learning algorithms of random forest and Naïve Bayes with the NIDS testing dataset to develop improved network security for ScottishGlen. Communicating the results can be achieved utilising confusion matrices and performing thorough testing.

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